BTS-IV-04.19-0303 Reg. No. [ ] A

B.Tech. Degree IV Semester Examination April 2019

EE 402 LOGIC DESIGN
(*2006 Scheme*)

Time: 3 Hours Maximum Marks: 100

PART A
(Answer *ALL* questions)

$(8 \times 5 = 40)$

I. (a) Convert the binary number 1101.101 to its decimal, octal and hexadecimal equivalent.
(b) State and prove DeMorgans theorem.
(c) Design full subtractor using NAND gates.
(d) Design monostable multivibrator using NOT gates.
(e) What is D FF (flip flop)? Convert JK FF into D FF.
(f) Briefly describe the types of memory.
(g) Explain tristate logic.
(h) With the help of a schematic, describe the interfacing of CMOS to TTL.

PART B

$(4 \times 15 = 60)$

II. (a) Simplify using Karnaugh map: (10)
(i) $f(A, B, C, D) = \sum m(1,2,3,4,5,7,8,9,11,12,13,15)$
(ii) $f(A, B, C, D) = \prod M(2,6,8,9,10,11,14)$
(b) What is a multiplexer? Design a 2 to 1 multiplexer using gates. (5)

OR

III. (a) Explain excess-3 code and grey code. Express 12_{10} in excess-3 code and grey code. (10)
(b) Write notes on error correction and detection codes. (5)

IV. (a) What is the disadvantage of ripple carry adder? Design a carry look ahead adder to eliminate this problem. (10)
(b) Multiply the binary numbers 1100 and 1111. (5)

OR

V. (a) What is full adder? Implement full adder circuit using (10)
(i) 4 to 1 multiplexer (ii) 3:8 decoder
(b) Design astable multivibrator using NAND gates for T = 1 ms. (5)

(P.T.O.)

VI.	(a)	What is sequential circuit? Draw the block diagram and explain.	(5)
	(b)	What is ring counter? Describe how the architecture of a ring counter differs from that of a Johnson counter. Design a 4 bit ring counter.	(10)

OR

VII.	(a)	Differentiate between synchronous and asynchronous inputs of flip flops.	(5)
	(b)	What is PAL? Differentiate between PAL and PLA. Realize $Y = A'B + AC$.	(10)
VIII.	(a)	Draw the circuit diagram of standard TTL connected as inverter and explain the working.	(10)
	(b)	Compare standard TTL and CMOS families on the basis of speed and power dissipation parameters.	(5)

OR

IX.	(a)	Why is ECL called non saturating logic?	(3)
	(b)	What is the main advantage accruing from this?	(2)
	(c)	With the help of a relevant circuit schematic describe the operation of ECL OR/NOR logic.	(10)

***